



Job Portal & Placement Site

Administrator Module

Project Team

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Introduction: The Problem

The traditional campus placement process remains fundamentally manual, creating significant operational challenges that impact both efficiency and accuracy.



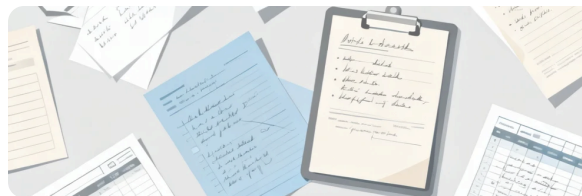
Time-Intensive Verification

Coordinators manually verify student eligibility criteria for each company, consuming valuable hours that could be spent on strategic activities.



Fragmented Data

Critical information is scattered across multiple spreadsheets, email threads, and physical documents with no single source of truth.



Error-Prone Operations

Manual data entry and processing creates operational bottlenecks and increases the risk of costly human errors in student placements.

Our Solution: Core Objectives

This administrator-side system transforms placement operations through intelligent automation, centralized data management, and powerful administrative tools designed to eliminate inefficiencies.

01 Centralize Student Data

Establish a unified repository consolidating all student profiles, academic records, and placement preferences into one secure, accessible location.

02 Automate Manual Tasks

Eliminate repetitive work by automating account initialization workflows and eligibility verification processes based on predefined criteria.

03 Streamline Job Posting

Enable efficient job opportunity creation with seamless Google Form integration for application collection and automated distribution.

04 Provide Master Sheet

Deliver comprehensive analytics through a single master sheet that logs all company interactions and tracks student application statuses in real-time.

05 Ensure Data Integrity

Implement admin-controlled profile modification protocols after initial student edits to maintain data accuracy and prevent unauthorized changes.



Technology Stack

A carefully selected, high-performance technology ecosystem engineered for scalability, security, and optimal user experience



Frontend Framework

1

ReactJS with Tailwind CSS for responsive, component-based UI development



Backend API

2

FastAPI (Python) delivering high-speed, asynchronous request handling



Database System

3

MongoDB providing flexible, document-based data storage



Authentication

JWT-based secure login system with token refresh mechanisms



Integration

Google APIs (Sheets & Forms) for seamless data synchronization



Version Control

GitHub for collaborative development and code management

Deployment Architecture

A sophisticated hybrid cloud infrastructure designed for reliability, performance, and seamless scalability across global user bases.



Frontend Layer

Deployed on Vercel with global CDN distribution and automated build pipelines for instant updates



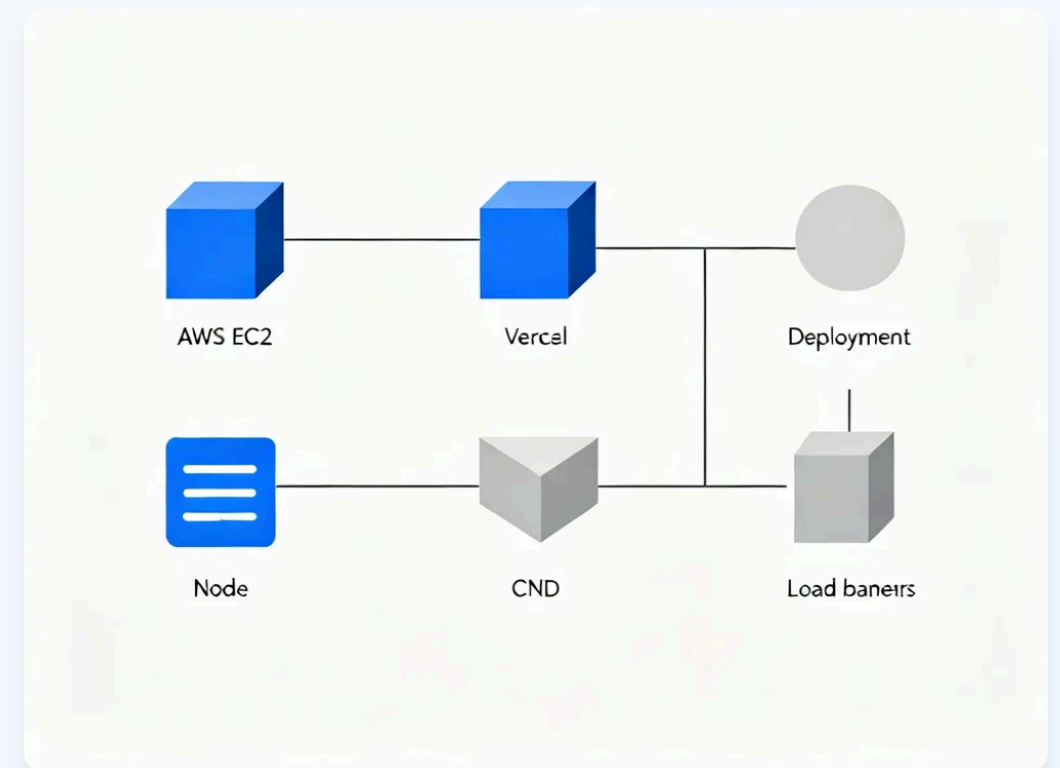
Backend Layer

Hosted on AWS EC2 with Dockerized FastAPI server and Nginx reverse proxy configuration



Security & Scale

HTTPS via AWS Certificate Manager with Elastic Load Balancer and Auto-Scaling Groups for dynamic capacity management

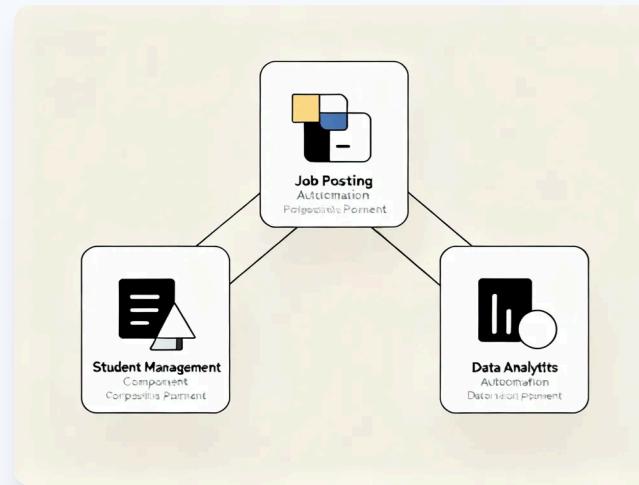


✓ Production-Ready Infrastructure

This deployment architecture ensures 99.9% uptime, automatic failover capabilities, and horizontal scaling to handle peak placement season traffic.

Core Admin Modules

The Administrator Module is built on three foundational pillars that work together to create a seamless placement management experience.



1

Student Management Module

Centralized control over student profiles, credentials, and data integrity



2

Job Posting & Google Form Automation

Automated job distribution with intelligent eligibility filtering



3

Master Data Sheet Module

Real-time application tracking via automated Google Sheets integration

Module 1: Student Management

This module provides comprehensive administrative control over student data throughout the entire placement lifecycle.

01 Initial Database Import

Bulk-create student accounts efficiently via CSV upload. Import hundreds of student records in seconds.

02 Account Initialization

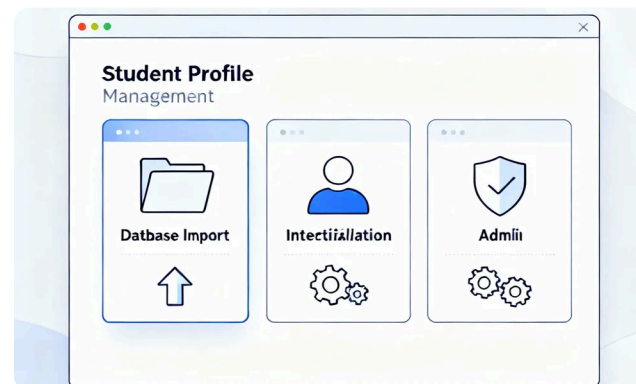
System automatically generates secure credentials and sends personalized welcome emails to each student.

03 Profile Data Control

Students can edit their profile once to ensure accuracy. After initial setup, all changes require admin approval to maintain data integrity.

04 Admin Override Access

Administrators maintain full access to directly modify any student profile when necessary, ensuring complete control.



Student Management Dashboard Interface

Module 2: Job Posting & Google Form Automation

Streamlines the creation of new job opportunities while ensuring only qualified students see relevant postings.

1 Job Creation

Admin defines job descriptions, eligibility criteria including CGPA thresholds, backlog limits, and branch requirements.

🎓 CGPA Thresholds

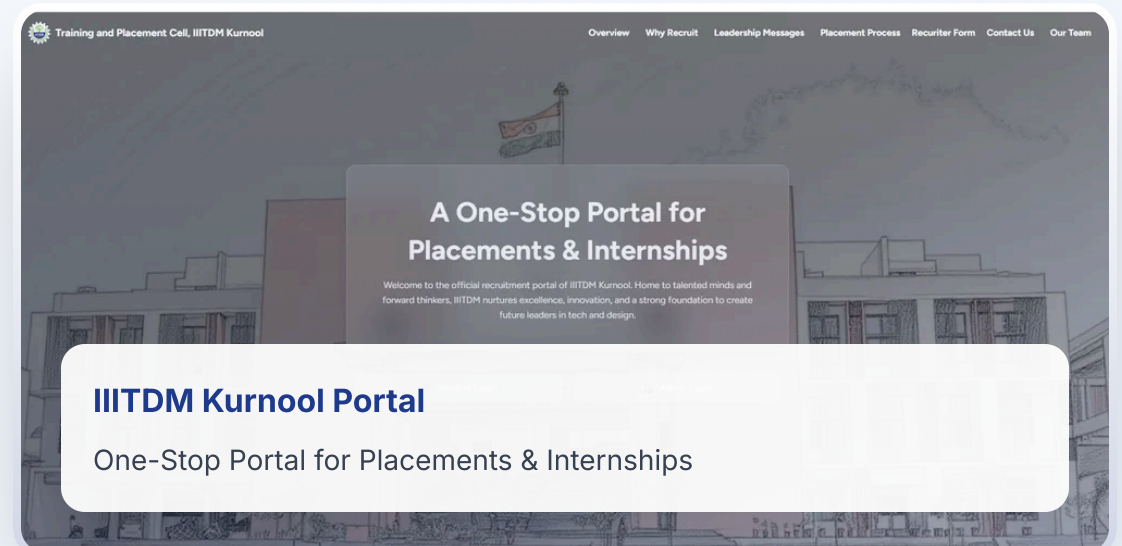
👤 Backlog Limits

🔑 Branch Requirements

2 Google Form Integration

Admin links application forms. The system automatically displays opportunities only to eligible students based on their profiles.

🔍 Smart Filtering 🔗 Form Integration 👥 Targeted Display



Intelligent Automation

Eliminates manual eligibility verification and ensures targeted job distribution

100%

Automated Filtering

0

Manual Verification

Real-time

Form Integration